

MINERALNYE VODY, Russian Federation

WMO: 370540

Lat: 44.225N

Lon: 43.082E

Elev: 321

StdP: 97.52

Time Zone: 3.00 (E03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-17.2	-13.1	-20.0	0.7	-17.1	-15.9	1.0	-12.9	11.1	-2.2	10.0	-3.5	3.6	100	0.547

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	13.4	34.7	21.0	32.9	20.4	31.1	20.0	22.5	31.0	21.7	29.8	21.0	28.8	4.5	100

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
19.9	15.2	26.2	19.1	14.4	25.1	18.2	13.6	24.2	68.0	31.1	65.0	30.0	62.1	28.7	26.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
(a)	(b)	(c)	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
10.3	9.0	8.1	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
			-20.7	37.0	5.2	2.2	-24.5	38.6	-27.5	39.9	-30.4	41.1	-34.2	42.6
			-20.9	24.2	4.9	1.1	-24.4	24.9	-27.3	25.6	-30.1	26.2	-33.7	26.9
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	10.7	-2.2	-1.0	4.3	10.4	16.0	20.5	23.3	23.2	17.8	11.1	4.5	-0.4
	DBStd	9.95	4.93	5.47	3.99	3.88	3.34	2.94	2.58	2.95	3.70	4.41	4.14	4.75
	HDD10.0	1445	379	309	182	41	3	0	0	0	1	40	168	322
	HDD18.3	3231	637	542	437	238	86	13	1	3	52	226	415	580
	CDD10.0	1692	0	0	4	53	188	314	413	408	234	73	3	0
	CDD18.3	437	0	0	0	1	13	77	156	153	36	2	0	0
	CDH23.3	4828	0	0	2	29	208	817	1715	1633	384	40	0	0
	CDH26.7	1991	0	0	0	4	38	285	781	756	116	10	0	0
Wind	WSAvg	3.5	3.6	3.8	4.1	4.0	3.4	3.3	3.2	3.3	3.3	3.5	3.5	3.5
Precipitation	PrecAvg	549	21	20	33	56	69	94	66	46	47	40	29	27
	PrecMax	685	41	39	68	153	113	251	184	108	119	90	65	56
	PrecMin	346	4	8	10	23	17	29	14	7	9	6	5	4
	PrecStd	86	10	8	14	27	26	47	36	28	26	22	16	13
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	11.0	15.1	21.8	26.6	29.8	34.1	36.9	36.9	33.2	28.1	19.1	13.0
		MCWB	6.4	8.9	12.7	15.3	18.7	20.0	22.0	21.4	20.2	17.3	12.1	8.3
	2%	DB	7.2	11.0	17.1	23.2	27.7	31.9	34.8	34.8	29.8	23.3	15.2	9.2
		MCWB	4.4	6.9	10.1	14.4	17.8	19.9	21.2	21.1	18.8	15.8	10.6	6.5
	5%	DB	4.9	7.7	14.0	21.0	25.8	29.9	32.9	32.9	27.3	20.8	12.2	7.0
		MCWB	2.9	4.8	8.6	13.2	17.4	19.8	20.6	20.4	17.9	14.7	9.1	5.1
	10%	DB	3.1	5.1	11.1	18.2	23.9	28.0	31.0	31.0	25.2	18.0	10.1	5.3
		MCWB	1.8	3.2	7.2	12.0	16.7	19.1	20.1	19.9	17.3	13.5	8.0	4.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	6.9	9.5	12.9	16.7	20.2	23.0	23.7	23.4	21.8	17.7	12.9	8.6
		MCDB	10.1	14.5	20.0	24.2	27.6	29.9	32.8	32.7	30.7	25.0	17.5	11.7
	2%	WB	4.7	6.8	10.6	15.0	18.8	21.5	22.6	22.4	19.8	16.3	11.1	6.7
		MCDB	7.0	10.5	16.0	21.8	25.6	28.7	31.2	31.1	26.5	22.0	14.7	8.7
	5%	WB	3.1	4.9	9.0	13.8	17.9	20.5	21.7	21.6	18.8	15.3	9.5	5.3
		MCDB	4.5	7.2	13.3	20.0	24.2	27.6	29.9	29.9	25.4	19.6	11.9	6.8
	10%	WB	1.8	3.3	7.5	12.5	16.9	19.7	21.0	20.8	17.9	14.0	8.2	4.1
		MCDB	2.8	4.7	10.7	17.5	22.8	26.3	28.9	28.7	23.9	17.6	9.9	5.2
Mean Daily Temperature Range	5% DB	MDBR	7.1	8.4	9.4	11.6	12.0	12.6	13.4	13.3	12.1	10.0	8.1	7.2
		MCDBR	10.8	13.1	16.5	17.0	16.0	16.2	16.9	16.6	16.3	15.7	13.6	10.9
		MCWBR	8.2	9.0	10.1	9.0	7.9	6.6	6.2	6.0	7.0	8.7	9.1	8.1
	5% WB	MCDBR	10.0	12.2	15.2	15.0	14.2	14.0	14.7	14.5	13.8	13.7	12.5	9.9
		MCWBR	7.7	8.6	9.7	8.4	7.4	6.6	6.1	6.1	6.6	7.8	8.6	7.5
Clear-Sky Solar Irradiance	taub	0.327	0.362	0.407	0.461	0.429	0.435	0.467	0.469	0.426	0.385	0.345	0.328	
	taud	2.317	2.232	2.148	2.035	2.215	2.244	2.165	2.157	2.247	2.319	2.372	2.322	
	Ebn at Noon	778	810	818	802	845	839	805	788	792	776	752	735	
	Edh at Noon	82	107	133	161	139	136	145	141	120	97	78	75	
All-Sky Solar Radiation	RadAvg	1.39	2.23	2.99	4.45	5.49	6.13	6.15	5.54	4.12	2.54	1.53	1.15	
	RadStd	0.25	0.44	0.33	0.51	0.57	0.37	0.40	0.34	0.46	0.35	0.22	0.15	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46)	Station Only	+0.36	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (9 neighbors)	+0.35	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+135	N/A

Nomenclature: See separate page